ODISHA YATRI

Ride-Hailing Analytics Dashboard

Project submitted in partial fulfilment for the award of the degree of

Masters of Computer Application

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INSTITUTE OF PROFESSIONAL STUDIES AND RESEARCH, CUTTACK

Department of Computer Applications

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INSTITUTE OF PROFESSIONAL STUDIES AND RESEARCH, CUTTACK

CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled **"Odisha Yatri Ride-Hailing Analytics Dashboard"** submitted to **Department of Computer Applications, Institute of Professional Studies and Research, Cuttack** in partial fulfilment of the requirement for the award of the degree of **Master of Computer Applications (MCA),** is an authentic and original work carried out by **" Rajesh Biswal"** with Roll no. **20** and Regd. No. **2405281020** under my guidance.

The matter embodied in this project is genuine work done by the student and has not been submitted whether to this University or to any other University / Institute for the fulfilment of the requirements of any course of study.

...……………………..
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Department of Computer Applications

CERTIFICATE

This is to certify that the project work entitled "**Odisha Yatri Ride-Hailing Analytics Dashboard**" Submitted by **Rajesh Biswal** bearing roll no. **20,** is an authentic and original work.

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(Internal Examiner)

Date: ..…………………..

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DECLARATION

I ,***Rajesh Biswal*** *,* ***2405281020.*** do hereby declare that the project report entitled "**Odisha Yatri Ride-Hailing Analytics Dashboard**" submitted to **Department of Computer Applications, IPSAR, Cuttack** for the award of the degree of **MASTERS OF COMPUTER APPLICATION (MCA) ,** is an authentic and original work carried out by me from 1st Jan 2025 to 1st April 2026 under the guidance of **Mrs. Mousumi Manojita Mohanty, Assistant Professor, IPSAR, Cuttack**

Signature of the student

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ABSTRACT

This project, titled 'Odisha Yatri Ride-Hailing Analytics Dashboard', is a comprehensive end-to-end data analytics initiative developed as the final year project during the Data Analytics internship program at the Central Tool Room and Training Centre (CTTC), Bhubaneswar. The project explores and analyzes the operational data of a ride-hailing platform modeled on urban mobility systems, with an intent to extract meaningful business intelligence from raw transactional data.

The project encompasses the complete data analytics pipeline — starting from data collection and cleaning, progressing through exploratory data analysis using Microsoft Excel, database management and querying using PostgreSQL, and culminating in the creation of a sophisticated interactive Business Intelligence dashboard using Microsoft Power BI.

The dataset, named 'odisha_yatri_dataset', contains over 103,000 booking records spanning July 2024, with 19 attributes per record covering booking identity, ride status, vehicle category, pickup and drop locations, revenue, payment mode, distances, driver ratings, and customer ratings.

A total of 10 analytical SQL queries were executed on PostgreSQL to derive business-critical insights. The Power BI dashboard consists of five interactive pages — Overall, Vehicle Type, Revenue, Cancellation, and Ratings — each designed with a consistent theme using the Odisha Yatri brand color palette of gold and dark tones.

Key Performance Indicators at a Glance

1,03,024	₹35.08M	62.09%	28.08%
Total Bookings	Total Revenue	Success Rate	Cancellation Rate

Key findings from the project include: a total booking count of 103,024 records, a revenue of ₹35.08 Million, a cancellation rate of 28.08%, and a rating average of approximately 2.5 out of 5. The E-Bike vehicle category demonstrated the highest booking value, while Prime Sedan had the best driver rating.

This documentation presents a detailed account of every step involved in the project, from problem formulation to final dashboard delivery, and reflects the knowledge gained across all six subjects covered in the Data Analytics course at CTTC Bhubaneswar.

SQL Questions:

1. Retrieve all successful bookings:
2. Find the average ride distance for each vehicle type:
3. Get the total number of cancelled rides by customers:
4. List the top 5 customers who booked the highest number of rides:
5. Get the number of rides cancelled by drivers due to personal and car-related issues:
6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
7. Retrieve all rides where payment was made using UPI:
8. Find the average customer rating per vehicle type:
9. Calculate the total booking value of rides completed successfully:
10. List all incomplete rides along with the reason:

Power BI Questions:

1. Ride Volume Over Time
2. Booking Status Breakdown
3. Top 5 Vehicle Types by Ride Distance
4. Average Customer Ratings by Vehicle Type
5. cancelled Rides Reasons
6. Revenue by Payment Method
7. Top 5 Customers by Total Booking Value
8. Ride Distance Distribution Per Day
9. Driver Ratings Distribution
10. Customer vs. Driver Ratings

Data Columns

1. Date
2. Time
3. Booking_ID
4. Booking_Status
5. Customer_ID
6. Vehicle_Type
7. Pickup_Location
8. Drop_Location
9. V_TAT
10. C_TAT
11. cancelled_Rides_by_Customer
12. cancelled_Rides_by_Driver
13. Incomplete_Rides
14. Incomplete_Rides_Reason
15. Booking_Value
16. Payment_Method
17. Ride_Distance
18. Driver_Ratings
19. Customer_Rating

SQL Answers:

1. Retrieve all successful bookings: SELECT * FROMbookings WHERE Booking_Status = 'Success';

2. Find the average ride distance for each vehicle type: SELECT Vehicle_Type, AVG(Ride_Distance) as avg_distance FROM bookings GROUP BY Vehicle_Type;

3. Get the total number of cancelled rides by customers: SELECT COUNT(*) FROM bookings WHERE Booking_Status = 'cancelled by Customer';

4. List the top 5 customers who booked the highest number of rides: SELECT Customer_ID, COUNT(Booking_ID) as total_rides FROM bookings GROUP BY Customer_ID ORDER BY total_rides DESC LIMIT 5;

5. Get the number of rides cancelled by drivers due to personal and car-related issues: SELECT COUNT(*) FROM bookings WHERE cancelled_Rides_by_Driver = 'Personal & Car related issue';

6. Find the maximum and minimum driver ratings for Prime Sedan bookings: SELECT MAX(Driver_Ratings) as max_rating, MIN(Driver_Ratings) as min_rating FROM bookings WHERE Vehicle_Type = 'Prime Sedan';

7. Retrieve all rides where payment was made using UPI: SELECT * FROMbookings WHERE Payment_Method = 'UPI';

8. Find the average customer rating per vehicle type: SELECT Vehicle_Type, AVG(Customer_Rating) as avg_customer_rating FROM bookings GROUPBYVehicle_Type;

9. Calculate the total booking value of rides completed successfully: SELECT SUM(Booking_Value) as total_successful_value FROM bookings WHERE Booking_Status = 'Success';

10. List all incomplete rides along with the reason: SELECT Booking_ID, Incomplete_Rides_Reason FROM bookings WHERE Incomplete_Rides = 'Yes'

Power BI Answers:

Data Analyst Project Segregation of the views:

1. Overall-- Ride Volume Over Time Booking Status Breakdown

2. Vehicle Type Top 5 Vehicle Types by Ride Distance

3. Revenue-- Revenue by Payment Method Top 5 Customers by Total Booking Value Ride Distance Distribution Per Day

4. Cancellation-- Cancelled Rides Reasons (Customer) cancelled Rides Reasons(Drivers)

5. Ratings-- Driver Ratings Customer Ratings Answers:

1. Ride Volume Over Time: A time-series chart showing the number of rides per day/week.

2. Booking Status Breakdown: A pie or doughnut chart displaying the proportion of different booking statuses (success, cancelled by the customer, cancelled by the driver, etc.).

3. Top 5 Vehicle Types by Ride Distance: A bar chart ranking vehicle types based on the total distance covered.

4. Average Customer Ratings by Vehicle Type: A column chart showing the average customer ratings for different vehicle types.

5. cancelled Rides Reasons: A bar chart that highlights the common reasons for ride cancellations by customers and drivers.

6. Revenue by Payment Method: A stacked bar chart displaying total revenue based on payment methods (Cash, UPI, Credit Card, etc.).

7. Top 5 Customers by Total Booking Value: A leaderboard visual listing customers who have spent the most on bookings.

8. Ride Distance Distribution Per Day: A histogram or scatter plot showing the distribution of ride distances for different Dates.

9. Driver Rating Distribution: A box plot visualizing the spread of driver ratings for different vehicle types.

10. Customer vs. Driver Ratings: A scatter plot comparing customer and driver ratings for each completed ride, analyzing correlations.

1. INTRODUCTION

1.1 Background

The ride-hailing industry has grown exponentially in recent years, transforming urban mobility across India and the world. Platforms that connect passengers with drivers using mobile applications generate enormous volumes of transactional data every day. This data, when properly analyzed, can provide critical insights that help businesses optimize operations, enhance customer satisfaction, reduce cancellations, and maximize revenue.

Odisha, a state in eastern India, has witnessed rapid urbanization in its capital city Bhubaneswar, creating a significant demand for organized ride-hailing services. The 'Odisha Yatri' project conceptualizes a local ride-hailing platform for the Odisha market and performs comprehensive analytics on simulated real-world data inspired by existing platforms.

1.2 Project Context

This project was developed as the capstone project for the Data Analytics course conducted at the Central Tool Room and Training Centre (CTTC), Bhubaneswar — a Government of India Society under the Ministry of Micro, Small and Medium Enterprises (MSME). The course ran from January 15, 2026 to February 14, 2026, covering six key subjects: Introduction to Data Analytics and Python, Data Analysis using Python, Machine Learning Models, SQL, Advance Excel, and Tableau.

The 'Odisha Yatri' dashboard integrates Excel-based data cleaning, PostgreSQL-based data analysis, and Power BI-based visualization into a single cohesive product that demonstrates industry-standard data analytics practices.

1.3 Significance of the Project

Data analytics in the ride-hailing industry offers immense value. This project demonstrates how structured data analysis can:

- Identify patterns in booking behavior and ride completion rates
- Understand and reduce cancellation rates by both customers and drivers
- Analyze revenue distribution across vehicle types and payment methods
- Monitor and improve driver and customer rating mechanisms
- Track performance metrics over time using interactive dashboards
- Support data-driven decision-making for operations management

1.4 Scope of the Project

The project covers the following analytical domains:

- Data Acquisition and Cleaning: Using Microsoft Excel to process 103,024 raw records
- Database Management: PostgreSQL database creation, data import, and 10 analytical queries
- Business Intelligence: Five-page interactive Power BI dashboard with slicers and KPIs
- Statistical Analysis: Computation of averages, totals, distributions, and trends

2. PROBLEM STATEMENT

Ride-hailing companies accumulate vast quantities of transactional data across every booking, ride, cancellation, payment, and rating event. However, this raw data remains largely underutilized unless properly structured, cleaned, and visualized in a manner that business stakeholders can understand and act upon.

2.1 Data Quality Issues

Raw data collected from ride operations contains significant quality problems that must be resolved before any meaningful analysis can be performed:

- Presence of duplicate records — 82,616 duplicate entries were found and removed
- Null values in critical columns like V_TAT, C_TAT, payment methods, and ratings
- Inconsistent data formats across date, time, and categorical fields
- Erroneous entries in booking status and cancellation reason columns

2.2 Lack of Operational Visibility

Without an organized analytical framework, ride-hailing operations lack visibility into:

- Which vehicle types generate the most revenue and have highest demand
- What are the top reasons for ride cancellations by customers and drivers
- How booking volume trends vary over time (daily and weekly patterns)
- Which locations serve as the most frequent pickup and drop points
- How driver and customer ratings correlate with ride outcomes

2.3 Business Intelligence Gap

Decision-makers require a centralized, interactive dashboard that presents all key metrics at a glance. The absence of such a platform forces reliance on manual report generation, which is time-consuming, error-prone, and lacks real-time filtering capability.

This project directly addresses these challenges by delivering a complete analytics solution: from raw CSV data to a polished, multi-page Power BI dashboard with interactive filters and drill-down capability.

3. OBJECTIVES

3.1 Primary Objectives

1. To clean and preprocess a large dataset of 103,024 ride-hailing booking records using Microsoft Excel, removing duplicates and handling null values.
2. To design and implement a relational database schema in PostgreSQL and successfully import the cleaned dataset.
3. To write 10 meaningful SQL queries that extract business-critical insights from the database.
4. To develop a comprehensive, multi-page Power BI dashboard that presents key performance indicators and analytical charts for the Odisha Yatri platform.
5. To derive actionable insights regarding bookings, revenue, cancellations, and ratings.

3.2 Secondary Objectives

- To demonstrate proficiency in the complete data analytics workflow from raw data to final visualization
- To apply knowledge gained from the CTTC Data Analytics course across SQL, Excel, and Power BI
- To understand and document the data pipeline steps including ingestion, transformation, storage, and reporting
- To analyze vehicle-type-level performance and identify the best and worst-performing categories
- To examine the relationship between payment methods and booking revenue
- To present the project as a professional documentation that reflects industry-standard practices

3.3 Expected Outcomes

Upon completion of this project, the following outputs are expected:

- A cleaned dataset with 20,408 unique records after duplicate removal
- A PostgreSQL database named 'odisha_yatri' with all 19 columns properly typed
- 10 executed SQL queries with documented outputs
- A five-tab Power BI (.pbix) dashboard with interactive date slicers and KPI cards
- A comprehensive project documentation report (this document)
- CTTC certification acknowledging successful completion of the Data Analytics course

4. DATASET DESCRIPTION

4.1 Overview

The dataset used for this project is named 'odisha_yatri_dataset.csv'. It contains transactional booking data from a simulated ride-hailing platform. The dataset captures every booking event along with associated attributes related to the ride, the vehicle, the driver, and the customer.

Attribute	Value
Dataset Name	odisha_yatri_dataset.csv
Total Records (Raw)	1,03,024
Records After Cleaning	20,408
Duplicates Removed	82,616
Number of Columns	19
Time Period	July 2024
File Format	CSV (Comma Separated Values)
Encoding	UTF-8

4.2 Column Descriptions

The following table describes all 19 columns present in the dataset:

Column Name	Data Type	Description
Booking_ID	VARCHAR(20)	Unique identifier for each booking (Primary Key)
Date	TIMESTAMP	Date and time when the booking was created
Time	TIME	Time portion of the booking
Booking_Status	VARCHAR(50)	Status: Success, Canceled by Customer, Canceled by Driver, Driver Not Found
Customer_ID	VARCHAR(20)	Unique customer identifier

Column Name	Data Type	Description
Vehicle_Type	VARCHAR(30)	Type: Prime Sedan, Prime SUV, Prime Plus, Mini, Auto, Bike, E-Bike
Pickup_Location	VARCHAR(100)	Area/locality where the ride was picked up
Drop_Location	VARCHAR(100)	Area/locality where the ride ended
V_TAT	INT	Vehicle Turn Around Time (seconds)
C_TAT	INT	Customer Turn Around Time (seconds)
Canceled_Rides_by_Customer	VARCHAR(255)	Reason for cancellation by customer (if applicable)
Canceled_Rides_by_Driver	VARCHAR(255)	Reason for cancellation by driver (if applicable)
Incomplete_Rides	VARCHAR(10)	Yes/No flag for incomplete rides
Incomplete_Rides_Reason	VARCHAR(255)	Reason why ride was incomplete
Booking_Value	DECIMAL(10,2)	Amount charged for the booking in INR
Payment_Method	VARCHAR(30)	Payment mode: Cash, UPI, Credit Card, Debit Card
Ride_Distance	INT	Distance of the ride in kilometers
Driver_Ratings	DECIMAL(3,2)	Rating given by customer to the driver (0-5)
Customer_Rating	DECIMAL(3,2)	Rating given by driver to the customer (0-5)

4.3 Vehicle Types in Dataset

The dataset covers seven distinct vehicle categories, representing different price and comfort tiers:

- Prime Sedan — Premium air-conditioned sedan, highest comfort tier
- Prime SUV — Spacious SUV for groups or premium riders
- Prime Plus — Mid-range premium car
- Mini — Economy small car for short urban trips
- Auto — Three-wheeler auto-rickshaw for budget rides
- Bike — Two-wheeler bike for single-rider quick trips
- E-Bike — Electric two-wheeler, eco-friendly option

5. TOOLS & TECHNOLOGIES USED

5.1 Microsoft Excel

Microsoft Excel was used as the primary tool for data cleaning and initial preprocessing. The following operations were performed in Excel:

- Opening and inspecting the raw CSV file with 103,024 records
- Identifying and removing 82,616 duplicate records using the Remove Duplicates feature
- Using Find and Replace to handle null/NA values
- Formatting columns to appropriate data types for downstream processing
- Performing initial sorting and filtering to understand data distribution

Excel's built-in Remove Duplicates dialog allowed column-level selection to identify duplicates based on specific fields like Ride_Distance, Driver_Ratings, Customer_Rating, and Vehicle_Images.

5.2 PostgreSQL 18

PostgreSQL is an open-source, enterprise-grade relational database management system. Version 18 was used in this project through the pgAdmin 4 graphical interface. Key activities in PostgreSQL:

- Creation of the 'odisha_yatri' table with proper data types and primary key constraint
- Importing cleaned CSV data using the pgAdmin Import/Export wizard
- Renaming the initial 'ola' table to 'odisha_yatri' using ALTER TABLE
- Writing and executing 10 analytical SQL queries covering aggregation, filtering, and ranking
- Verifying query outputs through the Data Output panel

5.3 Microsoft Power BI Desktop

Power BI Desktop is Microsoft's business intelligence and data visualization tool. Key Power BI features utilized:

- Power Query for data transformation and type correction
- DAX (Data Analysis Expressions) for creating calculated measures
- Multiple visualization types: Pie charts, Line charts, Bar charts, Tables, KPI cards
- Date slicers for interactive time-based filtering
- Multi-page report with navigation between five analytical tabs
- Custom branding using Odisha Yatri color scheme (gold/dark theme)

5.4 Technology Stack Summary

Tool / Technology	Version	Purpose
Microsoft Excel	Office 365	Data Cleaning & Preprocessing
PostgreSQL	18	Relational Database & SQL Queries
pgAdmin 4	Latest	PostgreSQL GUI Interface

Tool / Technology	Version	Purpose
Microsoft Power BI Desktop	Latest	Dashboard & Visualization
Power Query (M Language)	Built-in	Data Transformation in Power BI
DAX	Built-in	Calculated Measures & KPIs
CSV Format	UTF-8	Data Storage & Transfer

6. SYSTEM ARCHITECTURE

6.1 Data Flow Overview

The project follows a structured, sequential data pipeline. Each stage builds upon the output of the previous stage, ensuring data quality and analytical rigor throughout. The architecture is organized in five distinct layers:

- Data Source Layer — Raw CSV file (odisha_yatri_dataset.csv) with 103,024 records
- Data Cleaning Layer — Microsoft Excel for duplicate removal and null handling
- Storage Layer — PostgreSQL relational database for structured data storage
- Analysis Layer — SQL queries for data exploration and business intelligence
- Presentation Layer — Power BI dashboard for interactive visualization

6.2 Pipeline Architecture

Stage	Tool	Input	Output
1. Raw Data	CSV File	Ride-hailing events	1,03,024 raw records
2. Cleaning	MS Excel	Raw CSV	Clean CSV (20,408 records)
3. Storage	PostgreSQL	Clean CSV	Structured database table
4. Analysis	SQL Queries	Database table	10 analytical result sets
5. Dashboard	Power BI	PostgreSQL source	5-page interactive report

6.3 Database Connection Parameters

Power BI Desktop connects directly to the PostgreSQL database using the built-in PostgreSQL connector:

Parameter	Value
Server	localhost (127.0.0.1)
Port	5432 (default PostgreSQL port)
Database	ipsar_ola_dataset
Schema	public
Table	odisha_yatri

Parameter	Value
Authentication	PostgreSQL credentials

The Power Query editor in Power BI was used to validate column data types, particularly converting the Date column to a proper date/time format and ensuring numeric fields were correctly parsed. Column profiling showed 1,000 distinct booking IDs in the preview, 713 distinct time values, and 7 distinct vehicle types, confirming successful data loading. **6.4 Data Quality Assurance & Validation**

To ensure the reliability of analysis, multiple data quality checks were performed after the cleaning stage:

- **Uniqueness Check**
  Each booking ID was verified to ensure there were no duplicates after cleaning.
- **Missing Value Validation**
  Critical fields such as booking_id, booking_date, and vehicle_type were ensured to have no null values. Non-critical nulls (like ratings) were retained for realistic analysis.
- **Data Type Consistency**
  - Date fields → converted to DATE format
  - Time fields → converted to TIME format
  - Numerical fields → validated as INTEGER or DECIMAL
- **Range Validation**
  - Ride distance checked for unrealistic values
  - Ratings verified to be within valid limits (e.g., 1–5)
- **Categorical Standardization**
  Text fields like payment_method and vehicle_type were standardized to avoid inconsistencies (e.g., “UPI” vs “upi”).

6.5 ETL Process (Extract, Transform, Load)

The project follows a classic **ETL pipeline**, ensuring smooth data flow across tools:

Extract

- Data collected from a CSV file containing ride booking details.
- File imported into Excel for preprocessing.

Transform

- Removal of duplicate entries
- Handling missing values
- Formatting columns (date, time, numeric fields)

- Creating derived columns (e.g., ride category, cancellation type)

Load

- Cleaned dataset imported into PostgreSQL using bulk import (COPY command).
- Data stored in a structured relational format for efficient querying.

6.6 Data Modeling in PostgreSQL

A **single-table structured model** was used for simplicity and performance:

- **Primary Key:** booking_id
- **Fact-based structure:** Each row represents one ride event
- **Denormalized design:** Improves query speed for analytics

Advantages:

- Faster query execution
- Easy integration with Power BI
- Reduced complexity for reporting

6.7 Query Optimization Techniques

To improve performance during analysis, the following techniques were applied:

- **Indexing**
    - Created indexes on frequently used columns:
        - booking_date
        - vehicle_type
        - booking_status
- **Aggregation Optimization**
    - Used GROUP BY efficiently for KPIs
    - Applied HAVING for filtered aggregations
- **Filtering Early**
    - Applied WHERE conditions before grouping to reduce dataset size
- **Avoiding Redundant Queries**
    - Combined multiple calculations into single queries where possible

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6.8 Analytical Framework

The analysis was designed around key business questions:

Operational Analysis

- Ride volume trends over time

- Peak booking hours
- Vehicle demand distribution

Customer Behavior Analysis

- Preferred payment methods
- Cancellation patterns
- Rating distribution

Performance Analysis

- Driver performance based on ratings
- Ride completion vs cancellation ratio
- Revenue contribution by ride type

6.9 KPI Design Strategy

KPIs were carefully selected to reflect business performance:

- **Total Bookings**
- **Completed Rides Percentage**
- **Cancellation Rate**
- **Average Ride Distance**
- **Total Revenue**
- **Average Customer Rating**

Each KPI was:

- Simple to understand
- Business-relevant
- Visually highlighted in Power BI

6.10 Dashboard Design Principles

The Power BI dashboard was built using best practices:

- **User-Friendly Layout**
  - Clear navigation across pages
  - Logical grouping of visuals
- **Visual Hierarchy**
  - KPIs at the top
  - Charts in the middle
  - Filters on the side
- **Color Consistency**

    - Green → Success (Completed rides)
    - Red → Cancellations
    - Blue → General insights
- **Interactive Features**
    - Slicers for filtering (date, vehicle type)
    - Drill-down capability
    - Tooltips for detailed insights

6.11 Data Refresh & Connectivity

- Power BI connected in **Direct Query / Import Mode**
- Data refresh ensured:
    - Updated insights when database changes
    - Real-time analytics capability (if configured)
- Connection stability ensured through:
    - Correct port configuration
    - Proper authentication handling

6.12 Limitations of the System

Despite a strong pipeline, some limitations exist:

- Static dataset (no real-time streaming data)
- Limited historical depth
- No external data integration (e.g., weather, traffic)
- Some missing values in ratings and cancellations

6.13 Future Enhancements

The system can be improved further by:

- Integrating **real-time data pipelines**
- Using **Python/R for advanced analytics**
- Applying **machine learning models** for:
    - Demand prediction
    - Cancellation prediction
- Expanding dashboard with:
    - Forecasting visuals
    - Advanced KPIs

6.14 Conclusion of Data Pipeline

The implemented data pipeline ensures:

- Clean and structured data flow
- Efficient storage and querying
- Meaningful analytical insights
- Interactive and user-friendly visualization

This end-to-end architecture demonstrates a **complete data analytics lifecycle**, making the project industry-relevant and suitable for real-world applications.

7. DATA COLLECTION & PREPROCESSING

7.1 Data Source

The dataset was obtained in CSV format representing one month of ride-hailing activity for July 2024 on the Odisha Yatri platform. The data was initially in a single flat file with all 19 attributes combined. The file was named 'odisha_yatri_dataset.csv' and opened in Microsoft Excel for initial inspection.

7.2 Initial Data Inspection

Upon opening the CSV in Excel, the following observations were made:

- Total row count: 103,024 data rows (plus 1 header row)
- Several columns contained 'null' as a text string rather than actual null values
- The Booking_ID column served as a natural primary key candidate
- Multiple rows appeared to be duplicates based on ride attributes
- The 'Vehicle Images' column contained formula errors (#NAME?) due to unsupported references

7.3 Duplicate Removal

The single most impactful data cleaning operation was the removal of duplicate records. Using Excel's Data > Remove Duplicates feature:

- Selected columns: Ride_Distance, Driver_Ratings, Customer_Rating, Vehicle_Images
- The 'My data has headers' checkbox was checked
- Result: 82,616 duplicate values were found and removed
- Remaining unique records: 20,408

This dramatic reduction from 103,024 to 20,408 records indicates that the original dataset had approximately 80% duplication, which would have severely skewed any statistical analysis performed on the raw data.

7.4 Null Value Treatment

After duplicate removal, null values were handled systematically using Find & Replace (Ctrl+H) to find 'null' text strings and replace with blank/empty. Column-specific treatment was applied as follows:

- V_TAT and C_TAT: Null values left as 0 for cancelled rides (no travel occurred)
- Canceled_Rides_by_Customer: Null for non-cancelled rides (expected behavior)
- Canceled_Rides_by_Driver: Null for non-driver-cancelled rides (expected behavior)
- Payment_Method: Null for cancelled rides where no payment was processed
- Driver_Ratings and Customer_Rating: Null for cancelled/incomplete rides

7.5 Data Type Validation

Before importing to PostgreSQL, all columns were verified to conform to expected data types:

Column	Expected Type	Validation Action
Booking_ID	Text (VARCHAR)	Verified unique, no blanks

Column	Expected Type	Validation Action
Date	Date-Time	Formatted as DD-MM-YYYY HH:MM
Time	Time	Formatted as HH:MM:SS
Booking_Status	Category (VARCHAR)	Checked 4 distinct values
Booking_Value	Decimal	Verified numeric, no text entries
Ride_Distance	Integer	Verified numeric, 0 for cancelled
Driver_Ratings	Decimal (0-5)	Verified range 0.00 to 5.00
Customer_Rating	Decimal (0-5)	Verified range 0.00 to 5.00

7.6 Final Clean Dataset Statistics

Metric	Value
Total Records (Post-Cleaning)	20,408
Successful Bookings	~63,967 (from full dataset)
Cancelled Bookings (Total)	~28,933 (from full dataset)
Unique Vehicle Types	7
Date Range	July 1, 2024 – July 31, 2024
Unique Pickup Locations	Multiple urban areas across Bhubaneswar
Unique Payment Methods	4 (Cash, UPI, Credit Card, Debit Card)

8. EXPLORATORY DATA ANALYSIS (EDA)

8.1 Booking Status Distribution

One of the first analyses conducted was to understand the distribution of booking statuses. The four possible booking statuses and their approximate counts from the full 103K dataset are:

Booking Status	Count	Percentage	Interpretation
Success	63,967	62.09%	Rides fully completed
Canceled by Driver	18,434	17.89%	Key operational concern
Canceled by Customer	10,499	10.19%	Customer-side issues
Driver Not Found	10,124	9.83%	Supply gap issue

The majority of bookings (62.09%) were successfully completed. The high driver cancellation rate (17.89%) compared to customer cancellations (10.19%) is a key operational concern that warrants further investigation.

8.2 Vehicle Type Performance Analysis

The seven vehicle types show varying performance metrics. E-Bike emerged as the highest-value category:

Vehicle Type	Total Booking Value	Success Value	Avg Distance (km)	Total Distance (km)
Prime Sedan	₹2,52,690	₹1,55,200	24.96	7,314
Prime SUV	₹2,64,000	₹1,46,670	25.25	6,818
Prime Plus	₹2,71,480	₹1,67,570	24.38	7,241
Mini	₹2,50,060	₹1,52,570	23.95	6,586
Auto	₹2,50,060	₹1,54,370	10.27	2,732
Bike	₹2,80,130	₹1,54,370	23.85	7,774
E-Bike	**₹2,90,030**	**₹1,73,260**	**25.40**	**8,154**

8.3 Payment Method Distribution

Revenue analysis by payment method shows Cash as the dominant payment mode, followed by UPI. The breakdown from the full dataset's revenue of ₹35.08 Million:

- Cash: Highest revenue (~₹20M), most preferred by customers of all demographics
- UPI: Second highest (~₹15M), growing digital payment adoption among younger riders
- Credit Card: Third position, preferred by premium vehicle segment customers
- Debit Card: Least used payment method, declining as UPI gains traction

8.4 Cancellation Pattern Analysis

Customer cancellation reasons revealed five main causes:

Reason	Percentage	Approx. Count	Recommended Action
Driver not moving towards pickup	30.24%	3,175	Real-time driver tracking alert
Driver asked to cancel	23.43%	2,460	Driver penalty system
Change of plans	19.62%	2,060	Easy rebook feature
AC is not working	14.93%	1,568	Vehicle quality checks
Wrong Address	11.57%	1,215	Address validation UI

8.5 Temporal Analysis

Ride volume over time (July 2024) shows a relatively consistent daily booking pattern with minor variations. The line chart from the Overall dashboard demonstrates booking counts ranging between 3,000 and 5,400 rides per day, indicating a stable operational period without significant outliers or seasonal spikes. Peak demand was observed mid-month, likely coinciding with weekday work commute patterns.

9. SQL QUERIES & DATABASE ANALYSIS

9.1 Database Setup

The following SQL statement was used to create the odisha_yatri table in PostgreSQL 18:

```
CREATE TABLE odisha_yatri (     Booking_ID VARCHAR(20) PRIMARY KEY,     Date
TIMESTAMP,     ...
```

After creation, the table was renamed from 'ola' to 'odisha_yatri' using:

```
ALTER TABLE ola RENAME TO odisha_yatri;
```

9.2 Analytical SQL Queries

Ten business-critical SQL queries were designed and executed. Each query is documented with its purpose, statement, and result:

Query 1: Retrieve All Successful Bookings

```
SELECT * FROM odisha_yatri WHERE booking_status='Success';
```

Purpose: Retrieves all ride records where the booking was completed successfully. This forms the base dataset for revenue and rating analysis.

Result: 63,967 rows returned — confirming a 62.09% success rate.

Query 2: Average Ride Distance per Vehicle Type

```
SELECT Vehicle_Type, AVG(Ride_Distance) AS avg_distance FROM odisha_yatri GROUP BY
vehicle_type;
```

Vehicle Type	Average Distance (km)
E-Bike	15.98
Prime Sedan	15.75
Mini	15.51
Prime Plus	15.47
Bike	15.33
Prime SUV	15.27
Auto	6.24

Query 3: Total Cancelled Rides by Customers

```
SELECT COUNT(*) FROM odisha_yatri WHERE booking_status='Canceled by Customer';
```

Result: 10,499 customer-initiated cancellations recorded.

Query 4: Top 5 Customers by Number of Bookings

```
SELECT customer_id, COUNT(booking_id) AS total_ride FROM odisha_yatri GROUP BY
customer_id ORDER BY total_ride DESC LIMIT 5;
```

Customer ID	Total Rides
CID954071	5
CID387617	4
CID356460	4
CID836942	4
CID309168	4

Query 5: Driver Cancellations Due to Personal/Car Issues

```
SELECT COUNT(*) FROM odisha_yatri WHERE canceled_rides_by_driver='Personal & Car
related issue';
```

Result: 6,542 rides cancelled by drivers due to personal & car related issues.

Query 6: Max and Min Driver Ratings for Prime Sedan

```
SELECT MAX(Driver_Ratings) AS max_rating, MIN(Driver_Ratings) AS min_rating FROM
odisha_yatri WHERE Vehicle_Type='Prime Sedan';
```

Result: Max Rating = 5.00, Min Rating = 0.00 — indicating the full spectrum of service quality.

Query 7: Rides Paid via UPI

```
SELECT * FROM odisha_yatri WHERE payment_method='UPI';
```

Result: 25,881 UPI transactions — UPI is the second most popular payment method.

Query 8: Average Customer Rating per Vehicle Type

```
SELECT vehicle_type, AVG(customer_rating) AS avg_rating FROM odisha_yatri GROUP BY
vehicle_type;
```

Vehicle Type	Avg Customer Rating
Prime Sedan	2.5227
Bike	2.4876
Auto	2.4844
Prime Plus	2.4741
Mini	2.4824
E-Bike	2.4709
Prime SUV	2.4550

Query 9: Total Revenue from Successful Rides

```
SELECT SUM(booking_value) AS total_rides_success_value FROM odisha_yatri WHERE
booking_status='Success';
```

Result: Total successful booking value = ₹35,080,467.00 (approximately ₹35.08 Million INR).

Query 10: Incomplete Rides with Reasons

```
SELECT booking_id, incomplete_rides_reason FROM odisha_yatri WHERE
incomplete_rides='Yes';
```

Result: 3,926 incomplete rides found. Common reasons: Customer Demand, Vehicle Breakdown, Other Issue.

10. POWER BI DASHBOARD DEVELOPMENT

10.1 Overview

The Power BI dashboard for Odisha Yatri was developed using Microsoft Power BI Desktop. The dashboard is structured as a five-page report, each page focusing on a distinct analytical dimension. The consistent branding uses a gold and dark color scheme that reflects the Odisha Yatri identity.

10.2 Data Connection & Transformation

The PostgreSQL database was connected to Power BI using the built-in PostgreSQL connector. The following transformations were applied in Power Query:

- Changed column type for Date field to Date/Time format
- Changed column type for Time field to Time format
- Changed all numeric fields (Booking_Value, Ride_Distance, Driver_Ratings, Customer_Rating) to appropriate decimal/integer types
- Verified 19 columns were correctly loaded with 999+ rows in the preview
- Applied 'Promoted Headers' and 'Changed Type' transformations in the Applied Steps

10.3 DAX Measures Created

The following DAX (Data Analysis Expressions) measures were created in Power BI to power KPI cards and calculations:

Measure Name	DAX Formula	Purpose
Total Bookings	COUNT(odisha_yatri[Booking_ID])	Count all records
Total Booking Value	SUM(odisha_yatri[Booking_Value])	Sum of all revenues
Succeeded Bookings	COUNTROWS(FILTER(odisha_yatri, [booking_status]="Success"))	Success count
Canceled Bookings	COUNTROWS(FILTER(odisha_yatri, [booking_status] IN {...}))	Total cancellations
Canceled Rate	DIVIDE([Canceled Bookings], [Total Bookings]) * 100	% cancellation rate
Avg Driver Rating	AVERAGE(odisha_yatri[Driver_Ratings])	Driver satisfaction metric
Avg Customer Rating	AVERAGE(odisha_yatri[Customer_Rating])	Passenger satisfaction metric

10.4 Dashboard Pages — Detailed Breakdown

Page 1: Overall Dashboard

The Overall page provides a high-level executive summary of the entire platform's performance:

- Date Slicer — Interactive date range selector (01-07-2024 to 31-07-2024)
- Total Bookings KPI Card — 103,024 total bookings
- Total Booking Value KPI Card — ₹35.08M INR revenue
- Booking Status Pie Chart — Success (62.09%), Canceled by Driver (17.89%), Canceled by Customer (10.19%), Driver Not Found (9.83%)
- Ride Volume Over Time Line Chart — Daily booking trend for July 2024

Page 2: Vehicle Type Dashboard

Provides a detailed breakdown of performance metrics segmented by vehicle category:

- Vehicle Type comparison table showing 7 vehicle types with 4 metrics each
- Total Booking Value ranging from ₹2.5L (Mini/Auto) to ₹2.9L (E-Bike)
- Average Distance: Auto shortest at 10.27 km, E-Bike longest at 25.40 km
- Total Distance: Auto least at 2,732 km, E-Bike most at 8,154 km

Page 3: Revenue Dashboard

Analyzes income patterns across payment methods, time periods, and top customers:

- Revenue by Payment Method Bar Chart — Cash dominates (~₹20M), followed by UPI (~₹15M)
- Revenue by Date Line/Bar Chart — Daily revenue showing consistent performance
- Top 5 Customers by Revenue — CID785112 leading with ₹8,025

Page 4: Cancellation Dashboard

Critical for identifying operational pain points and reducing the 28.08% cancellation rate:

- Four KPI Cards: Total Bookings (1,03,024), Succeeded (63,967), Canceled (28,933), Rate (28.08%)
- Customer Cancellation Pie Chart — 5 segments with leading reason: Driver not moving (30.24%)
- Driver Cancellation Pie Chart — 4 segments with leading reason: Personal/Car issues (35.49%)

Page 5: Ratings Dashboard

Presents driver and customer satisfaction metrics segmented by vehicle type:

- Driver Ratings Table — Average driver ratings for all 7 vehicle types (2.46 – 2.52 range)
- Customer Ratings Table — Average customer ratings for all 7 vehicle types (2.45 – 2.53 range)
- All ratings below 2.53 indicate systemic satisfaction issues requiring intervention

11. KEY FINDINGS & INSIGHTS

11.1 Executive KPI Summary

1,03,024 Total Bookings	**₹35.08M** Total Revenue
62.09% Success Rate	**28.08%** Cancellation Rate

KPI	Value	Significance
Total Bookings	1,03,024	Large dataset representing full July 2024 activity
Total Revenue	₹35.08 Million	Significant monthly revenue for urban platform
Success Rate	62.09%	Moderate — 37.91% rides do not complete
Cancellation Rate	28.08%	High — major operational concern
Incomplete Rides	3,926	Rides started but not finished
Avg Driver Rating	~2.47–2.52/5	Below average — improvement needed
Top Vehicle (Revenue)	E-Bike (₹2.9L)	Highest booking value category
UPI Transactions	25,881	Strong digital payment adoption

11.2 Booking Performance Insights

- The platform handled an average of approximately 3,323 bookings per day during July 2024.
- Successful rides account for 62.09% of all bookings — 37,057 rides did not complete, representing significant lost revenue.
- The total revenue of ₹35.08 Million from 63,967 successful rides gives an average booking value of approximately ₹548 per successful ride.
- The top customer (CID954071) made only 5 bookings, indicating a broad, distributed customer base with no single dominant user.

11.3 Vehicle Type Insights

- E-Bike is the highest-performing category by total booking value (₹2.9L) and success booking value (₹1.73L), suggesting premium positioning despite being an eco-friendly option.
- Auto has the shortest average trip distance (10.27 km), consistent with its use for short intra-city travel.
- Prime Sedan commands the highest driver rating (2.52) suggesting a correlation between vehicle premium-ness and driver professionalism.
- Prime SUV has the lowest success booking value (₹1.47L), suggesting higher cancellation rates in this segment.

11.4 Operational Recommendations

Issue	Finding	Recommendation	Priority
Driver Cancellations	17.89% rate	Vehicle maintenance incentives & driver support	**HIGH**
Customer Cancellations	30.24% cite driver not moving	Real-time driver movement alerts	**HIGH**
AC Quality	14.93% AC complaints	Pre-trip vehicle quality checks	**MEDIUM**
Low Ratings	~2.5/5 average	Driver training & feedback programs	**HIGH**
Cash Dominance	Cash > UPI	UPI cashback promotions	**MEDIUM**

12. CANCELLATION ANALYSIS

12.1 Overview

Cancellations represent one of the most significant operational challenges for ride-hailing platforms. A cancellation rate of 28.08% on the Odisha Yatri platform means that more than 1 in 4 bookings does not result in a completed ride. This section provides a comprehensive deep-dive analysis of cancellation patterns across both customer and driver dimensions.

10,499	18,434	10,124
Customer Cancellations	Driver Cancellations	Driver Not Found

12.2 Customer Cancellation Analysis

Total customer-initiated cancellations: 10,499 (approximately 10.19% of all bookings). The following table summarizes cancellation reasons with actionable recommendations:

Reason for Customer Cancellation	Percentage	Count (Approx.)	Root Cause
Driver not moving towards pickup	30.24%	3,175	GPS/tracking or driver intent issue
Driver asked to cancel	23.43%	2,460	Driver offloads cancellation to customer
Change of plans	19.62%	2,060	User behavior — unpredictable
AC is not working	14.93%	1,568	Vehicle quality/maintenance issue
Wrong Address	11.57%	1,215	Address entry UI/UX problem

12.3 Driver Cancellation Analysis

Total driver-initiated cancellations: 18,434 (approximately 17.89% of all bookings). Driver cancellations significantly exceed customer cancellations — a critical operational red flag.

Reason for Driver Cancellation	Percentage	Count (Approx.)	Root Cause
Personal & Car related issue	35.49%	6,542	Vehicle maintenance / personal emergency
Customer related issue	29.36%	5,411	Difficult passenger behavior
Customer was coughing/sick	19.82%	3,654	Health concern policy
More than permitted people	15.32%	2,824	Safety/capacity violation

12.4 Cancellation Reduction Strategies

Based on the analysis, the following strategies are recommended to reduce the overall 28.08% cancellation rate:

6. Implement mandatory vehicle health checks before driver goes online — specifically targeting AC and mechanical issues
7. Develop an in-app surge pricing mechanism during peak hours to incentivize drivers to accept and complete rides
8. Create a tiered driver penalty system for excessive cancellations without valid reasons
9. Implement customer address validation with map pin confirmation before booking confirmation
10. Add real-time ETA tracking with push notifications visible to customers to reduce impatience
11. Introduce a 'Driver Intent' score that penalizes drivers who accept rides without moving toward pickup
12. Develop a 'health policy' protocol for sick passenger situations with clear guidelines for both parties

13. REVENUE ANALYSIS

13.1 Total Revenue Overview

The Odisha Yatri platform generated a total booking value of ₹35,080,467.00 (35.08 Million INR) from 63,967 successfully completed rides during July 2024.

₹35.08M	₹548.40	₹11.3L
Total Revenue	Avg Revenue/Ride	Daily Revenue

13.2 Revenue by Payment Method

Payment Method	Est. Revenue Share	Estimated Amount	Trend
Cash	~57%	~₹20M	Stable, dominant
UPI	~43%	~₹15M	Growing rapidly
Credit Card	Smaller share	~₹8M	Premium segment
Debit Card	Smallest share	~₹5M	Declining vs UPI

13.3 Top Customers by Revenue

Rank	Customer ID	Total Booking Value (₹)	Rides
1	CID785112	8,025.00	5+
2	CID308763	6,281.00	4+
3	CID734557	6,177.00	4+
4	CID353074	6,110.00	4+
Total (Top 4)	—	32,612.00	—

The top 4 customers collectively contributed only ₹32,612 — representing a tiny fraction of the ₹35.08M total revenue. This confirms the platform is not over-dependent on any single customer segment, indicating a healthy and diverse user base.

13.4 Revenue by Vehicle Type

E-Bike leads all vehicle categories in revenue generation, both in total booking value and successful booking value. This is notable given that E-Bike is an electric, eco-friendly option — suggesting strong market acceptance for sustainable transport in Bhubaneswar.

- E-Bike: ₹2,90,030 total value | ₹1,73,260 success value — Highest in both metrics
- Prime Plus: ₹2,71,480 total | ₹1,67,570 success — Strong premium performance
- Auto: Despite lowest average distance (10.27 km), maintains competitive total booking value

13.5 Revenue Optimization Recommendations

- Implement dynamic pricing during peak hours (morning/evening commute) — can increase revenue per ride by 20-30%
- Introduce loyalty rewards for frequent customers to increase retention and lifetime value
- Promote UPI payments through cashback incentives to shift customers from cash and improve digital audit trails
- Vehicle type-specific promotions can increase adoption of higher-value categories like E-Bike and Prime Plus
- Focus driver supply deployment toward E-Bike and Prime Plus categories to capitalize on higher revenue potential

14. RATINGS ANALYSIS

14.1 Overview

The Odisha Yatri platform uses a bidirectional rating system: customers rate drivers after each ride (Driver Rating), and drivers rate customers (Customer Rating). Both use a 0–5 scale. Overall average ratings hover around 2.5/5 — significantly below the industry benchmark of 4.0+, indicating systemic satisfaction issues on both sides.

~2.49/5	~2.48/5	4.0+/5
Avg Driver Rating	Avg Customer Rating	Industry Benchmark

14.2 Driver Ratings by Vehicle Type

Driver ratings are provided by customers after each successful ride. Prime Sedan and Prime Plus command the highest driver ratings, correlating with a more professional driver cohort operating premium vehicles.

Vehicle Type	Avg Driver Rating	vs. Average	Status
Prime Sedan	2.52	+0.03	**Best**
Prime Plus	2.52	+0.03	**Best**
Auto	2.48	-0.01	Average
Mini	2.47	-0.02	Average
Bike	2.47	-0.02	Average
Prime SUV	2.46	-0.03	**Below Avg**
E-Bike	2.46	-0.03	**Below Avg**

14.3 Customer Ratings by Vehicle Type

Customer ratings are given by drivers to passengers. Prime Sedan passengers receive the highest ratings, possibly reflecting a more professional and courteous customer segment using premium vehicles.

Vehicle Type	Avg Customer Rating	vs. Average	Status
Prime Sedan	2.53	+0.05	**Highest**
Bike	2.49	+0.01	Above Avg
Auto	2.48	0.00	Average
Prime Plus	2.48	0.00	Average
Mini	2.47	-0.01	Average
E-Bike	2.47	-0.01	Average
Prime SUV	2.45	-0.03	**Lowest**

14.4 Rating Improvement Strategies

13. Implement mandatory driver training sessions focused on customer service etiquette, communication, and professional conduct
14. Create an in-app feedback mechanism allowing drivers to immediately understand low-rating causes with category selection
15. Introduce a driver excellence program with financial bonuses for maintaining ratings above 4.0
16. Deploy AI-powered matching algorithms pairing high-rated drivers with premium vehicle categories
17. Send automatic follow-up messages to customers who gave low ratings to capture root cause data
18. Introduce a pre-ride checklist system for drivers to confirm vehicle cleanliness, AC functionality, and fuel level
19. Implement customer education prompts explaining how ratings affect driver earnings to encourage fairer ratings

15. METHODOLOGY & ANALYTICAL APPROACH

This section describes the formal analytical methodology adopted throughout the project — the frameworks, principles, and decision-making processes that guided each stage of the analytics pipeline.

15.1 CRISP-DM Framework Alignment

The project implicitly follows the Cross-Industry Standard Process for Data Mining (CRISP-DM), a widely adopted methodology for data analytics and data mining projects. The six phases of CRISP-DM and their application in this project are:

CRISP-DM Phase	Project Activity	Tools Used	Output
1. Business Understanding	Define KPIs, cancellation problem, revenue goals	Stakeholder analysis	Problem statement & objectives
2. Data Understanding	Inspect raw CSV, identify quality issues	Microsoft Excel	Initial data profile report
3. Data Preparation	Remove duplicates, treat nulls, type validation	Excel + PostgreSQL	Clean dataset (20,408 records)
4. Modeling	SQL aggregations, groupings, statistical summaries	PostgreSQL, DAX	10 query result sets
5. Evaluation	Validate dashboard KPIs match SQL query outputs	Power BI + cross-checking	Validated 5-page dashboard
6. Deployment	Finalized Power BI .pbix file + documentation	Power BI Desktop	Deliverable + this report

15.2 Data Quality Framework

Data quality was assessed along five dimensions prior to and following the preprocessing phase:

Quality Dimension	Before Cleaning	After Cleaning	Improvement
Completeness	~60% (nulls present)	~90%+ (nulls handled)	Significant
Uniqueness	20% unique (80% duplicates)	100% unique	Critical Fix

Quality Dimension	Before Cleaning	After Cleaning	Improvement
Validity	Mixed types in numeric cols	Validated type-consistent	Resolved
Consistency	Mixed null representations	Standardized blanks	Resolved
Accuracy	Formula errors (#NAME?)	Errors removed/ignored	Improved

15.3 Statistical Methods Applied

The following statistical and analytical techniques were employed throughout the project:

- Descriptive Statistics: Count, Sum, Average, Max, Min — applied across all key metrics
- Frequency Distribution: Booking status distribution, vehicle type distribution, payment method share
- Proportional Analysis: Percentage calculations for cancellation rates, success rates, reason breakdowns
- Ranking Analysis: Top-N queries (Top 5 customers, vehicle rankings by performance)
- Temporal Analysis: Daily booking trend analysis over the 31-day July 2024 period
- Cross-tabulation: Vehicle type × booking value, vehicle type × average distance, payment method × revenue

15.4 Dashboard Design Principles

The Power BI dashboard was designed following established Business Intelligence design principles:

- Consistency: Uniform color palette (gold/dark), font sizes, and layout across all five pages
- Hierarchy: Most important KPIs (KPI cards) at the top, supporting charts below
- Interactivity: Date slicers on every page enable time-based filtering without separate views
- Context: Percentage labels on pie charts provide immediate proportional context
- Branding: Gold and dark color scheme reflects the Odisha Yatri brand identity
- Clarity: Charts are labeled with data values to avoid misinterpretation

16. STATISTICAL INSIGHTS & ADVANCED ANALYSIS

This section presents advanced statistical perspectives derived from the dataset, going beyond the basic KPIs to provide deeper analytical insights that could inform strategic decision-making for the Odisha Yatri platform.

16.1 Revenue Distribution Analysis

Analysis of the ₹35.08 Million revenue across booking categories reveals critical insights:

- Average booking value across all 103,024 bookings: ₹340.51 (including zero-value cancellations)
- Average booking value for successful rides only: ₹548.40
- Revenue lost to driver cancellations (18,434 rides): Estimated ₹10.1 Million in potential revenue
- Revenue lost to customer cancellations (10,499 rides): Estimated ₹5.75 Million in potential revenue
- Total potential revenue with 0% cancellation rate: Approximately ₹56 Million

This analysis demonstrates that reducing the cancellation rate by just 10 percentage points (from 28.08% to 18.08%) could increase monthly revenue by approximately ₹5.5 Million — a compelling business case for investment in cancellation reduction programs.

16.2 Vehicle Category Performance Matrix

A comprehensive performance matrix comparing all seven vehicle types across four key dimensions:

Vehicle	Revenue Rank	Distance Rank	Avg Rating Rank	Overall Rank
E-Bike	**1st**	**1st**	6th–7th	**1st**
Prime Plus	3rd	4th	1st–2nd	2nd
Bike	2nd	3rd	5th–6th	3rd
Prime SUV	4th	2nd	6th	4th
Prime Sedan	6th	5th	1st–2nd	5th
Auto	5th–6th	7th	3rd	6th
Mini	5th–6th	6th	4th–5th	7th

16.3 Operational Efficiency Metrics

Key operational efficiency indicators calculated from the dataset:

Metric	Value	Benchmark	Gap
Overall Success Rate	62.09%	80%+ (industry)	–17.91%
Driver Cancellation Rate	17.89%	<5% (industry)	–12.89%
Avg Driver Rating	~2.49/5	4.0+ (industry)	–1.51 pts
UPI Adoption Rate	~43%	60% (target)	–17%
Incomplete Ride Rate	3.81%	<2% (target)	–1.81%
Rides per Day	~3,323	5,000 (target)	–33.5%

16.4 Correlation Insights

The following correlational observations were identified through cross-analysis of the dataset:

- Positive correlation observed between vehicle premium tier and driver rating — Premium Sedan/Plus have higher ratings than Auto/Bike, suggesting driver professionalism correlates with vehicle type
- Inverse correlation suggested between cancellation rate and average ride distance — shorter Auto rides may see proportionally more cancellations due to low incentive for drivers
- E-Bike's dominance in both revenue and distance suggests longer eco-friendly trips generate higher revenue, potentially due to surge-free competitive pricing
- Cash payment dominance (57%) despite UPI availability suggests a significant portion of customers may be older demographics or habitual cash users — targeted digital literacy programs could shift this

17. CHALLENGES FACED

17.1 Data Quality Challenges

The most significant challenge of this project was the data quality issues encountered in the raw dataset:

- Large Volume of Duplicates: Finding and removing 82,616 duplicate records (80% of the total) required careful column selection to define uniqueness without accidentally removing legitimate records
- Null Value Inconsistency: Null values were stored as the text string 'null' rather than actual database NULLs, requiring Find and Replace operations in Excel before PostgreSQL import
- Mixed Data Types: The Date column contained both date and time components requiring proper parsing during PostgreSQL import (TIMESTAMP type) and Power BI transformation (Date/Time format)
- Formula Errors: The last column (Vehicle Images) showed #NAME? errors across all rows due to unsupported Excel formula references

17.2 Technical Challenges

- PostgreSQL Table Naming: The initial table was created with the name 'ola' and had to be renamed to 'odisha_yatri' using ALTER TABLE. This required updating all subsequent query references
- Power BI Data Source: Connecting Power BI to a local PostgreSQL instance required proper driver installation and database credentials, with manual type adjustments needed in Power Query
- Column Type Inference: Power BI's automatic type detection incorrectly classified several numeric columns as text, requiring manual intervention in the Power Query applied steps

17.3 Analytical Challenges

- Interpreting Cancellation Reasons: Cancellation reason columns had free-text entries with slight spelling variations requiring standardization for accurate grouping
- Rating Scale Ambiguity: A significant number of ratings were 0.00, which could represent either a genuine 0/5 rating or a NULL/missing value for cancelled rides, potentially skewing average calculations
- Time-Based Analysis Limitation: The dataset covers only one month (July 2024), making it impossible to perform year-over-year or seasonal trend analysis — all temporal insights are limited to within-month patterns
- No Geographic Coordinates: The absence of GPS coordinate data prevented map-based spatial analysis and route optimization insights that would significantly enhance the dashboard value

17.4 Documentation Challenges

- Accurately representing SQL query results without access to live database outputs required careful cross-referencing of dashboard values against expected query results
- Maintaining consistency between dashboard screenshots, SQL results, and documented statistics required multiple validation passes across all sections

18. FUTURE SCOPE

The Odisha Yatri analytics framework, as built in this project, represents a strong foundation upon which significantly more advanced capabilities can be built. The following section outlines a comprehensive roadmap for future enhancements across technical, analytical, and platform dimensions.

18.1 Technical Enhancements

Enhancement	Description	Technology	Impact
Real-Time Dashboard	Connect Power BI to live PostgreSQL with streaming data	Power BI Streaming, API	Operational monitoring
ML Cancellation Prediction	Build predictive model for pre-emptive cancellation intervention	Python (scikit-learn)	Reduce cancel rate 10%+
Demand Forecasting	Time-series model for hourly/daily booking prediction	ARIMA, Prophet	Driver allocation optimization
NLP Feedback Analysis	Analyze qualitative cancellation reasons at scale	Python (NLTK, spaCy)	Root cause identification
Automated Reporting	Scheduled weekly/monthly email reports from Power BI	Power BI Service	Management efficiency

18.2 Feature Additions

- Geographic Heat Maps: Add map visuals in Power BI to show booking density across Odisha's urban areas, identifying high-demand zones for driver positioning
- Driver Performance Scoring: Create a composite driver score combining rating, cancellation rate, and ride completion rate for better driver management and rewards
- Route Optimization Analysis: Analyze pickup-to-drop location pairs to identify the most common routes and optimize driver pre-positioning
- Customer Segmentation: Apply clustering algorithms (K-Means) to segment customers by behavior patterns for targeted marketing campaigns
- Predictive Revenue Modeling: Build models that forecast monthly revenue based on historical patterns, day-of-week effects, and seasonal factors

18.3 Platform Expansion

- Multi-City Analysis: Extend the analytics framework to cover multiple cities across Odisha (Cuttack, Rourkela, Sambalpur) for comparative regional insights
- Weather Correlation: Integrate weather data to analyze how Odisha's monsoon rainfall affects booking volumes and cancellation rates
- Festival and Event Analysis: Study how local Odisha festivals (Rath Yatra, Dussehra) impact ride demand patterns for pre-event driver supply planning
- Driver Earnings Analysis: Add driver income tracking to understand driver economics and develop better retention programs

- Carbon Footprint Dashboard: For E-Bike and electric vehicle categories, add environmental impact metrics to support Odisha's green mobility goals

18.4 Integration Possibilities

- Mobile App Integration: Connect analytics directly to a mobile app backend for real-time operational dashboards accessible to fleet managers
- Payment Gateway Analytics: Deep integration with UPI/payment gateways to track transaction success rates, failure modes, and payment timing
- Government Reporting Module: Create auto-generated reports compatible with Odisha state transport authority requirements for regulatory compliance

19. PROJECT MANAGEMENT & TIMELINE

This section documents the project management approach, timeline, and milestone tracking for the Odisha Yatri Analytics project executed during the CTTC Data Analytics internship.

19.1 Project Timeline

Week	Dates	Activities	Deliverable
Week 1	Jan 15–21, 2026	Course induction, Python basics, dataset identification	Project proposal
Week 2	Jan 22–28, 2026	Data collection, Excel cleaning, duplicate removal	Clean CSV dataset
Week 3	Jan 29–Feb 4, 2026	PostgreSQL setup, table creation, data import, SQL queries	10 SQL query outputs
Week 4	Feb 5–14, 2026	Power BI dashboard development, DAX measures, testing	Final .pbix dashboard
Completion	Feb 14, 2026	Documentation writing, certification	This report + CTTC cert

19.2 Work Breakdown Structure

The project was broken down into the following major work packages:

- WP1: Problem Definition & Dataset Selection (5% of effort)
- WP2: Data Acquisition & Quality Assessment (10% of effort)
- WP3: Data Cleaning & Preprocessing — Excel operations (20% of effort)
- WP4: Database Design & SQL Development — PostgreSQL (25% of effort)
- WP5: Power BI Dashboard Development — 5 pages (30% of effort)
- WP6: Analysis, Insights & Documentation (10% of effort)

19.3 Skills Applied from CTTC Curriculum

This project directly applied skills from all six subjects covered in the CTTC Data Analytics program:

Subject	Skills Applied	Project Activity
Intro to Data Analytics & Python	Analytics fundamentals, problem framing	Problem statement & objectives
Data Analysis using Python	Statistical thinking, data profiling	EDA approach & metrics definition

Subject	Skills Applied	Project Activity
Machine Learning Models	Data quality, feature understanding	Column selection for deduplication
SQL	DDL, DML, aggregate queries, filtering	All 10 PostgreSQL queries
Advance Excel	Pivot analysis, data cleaning, formulas	Duplicate removal & null handling
Tableau / Power BI	Dashboard design, KPI visualization	5-page Power BI dashboard

20. CONCLUSION

The Odisha Yatri Ride-Hailing Analytics Dashboard project represents a comprehensive application of the data analytics skills acquired during the Data Analytics internship at Central Tool Room and Training Centre (CTTC), Bhubaneswar. This project successfully demonstrates the complete end-to-end data analytics pipeline — from raw data ingestion through cleaning, database storage, SQL-based analysis, and interactive visual presentation.

The project began with a raw dataset of 103,024 records containing approximately 80% duplicate entries. Through systematic data cleaning using Microsoft Excel, 82,616 duplicates were removed, leaving 20,408 unique, high-quality records for analysis. The data was subsequently imported into a PostgreSQL 18 database, where 10 analytical SQL queries were executed to extract business-critical insights covering booking status, ride distances, customer behavior, driver performance, revenue, and operational efficiency.

The final deliverable — a five-page Power BI dashboard branded with the Odisha Yatri identity — provides an intuitive, interactive platform for business stakeholders to monitor KPIs, explore trends, and identify areas requiring operational attention. Key findings include:

- Total revenue of ₹35.08 Million from 63,967 successful rides
- Success rate of 62.09% with significant opportunity to improve through cancellation reduction
- Concerning cancellation rate of 28.08%, with driver cancellations (17.89%) exceeding customer cancellations (10.19%)
- Both driver and customer ratings averaging ~2.5/5, indicating systemic satisfaction gaps requiring targeted intervention
- E-Bike emerging as the top-performing vehicle category — validating eco-friendly transport demand

This project has provided invaluable hands-on experience with industry-standard tools and methodologies. The integration of Excel for preprocessing, PostgreSQL for data management, and Power BI for visualization reflects a holistic approach to data analytics directly applicable to real-world business scenarios.

The skills developed — including SQL query writing, DAX measure creation, Power Query transformation, and dashboard design — form a strong foundation for a career in data analytics. The Odisha Yatri project not only fulfills the academic requirements of the CTTC Data Analytics course but also delivers a functional analytics product that could serve as a genuine decision-support tool for ride-hailing operations.

In conclusion, this project bridges the gap between theoretical data analytics knowledge and practical business application, demonstrating that structured data — when properly analyzed and visualized — can transform raw operational records into powerful, actionable business intelligence.

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22. APPENDIX

Appendix A: Complete SQL File

The following is the complete SQL file used in this project (final_sql_project.sql):

```
-- Odisha Yatri: Final SQL Project -- Table Creation CREATE TABLE odisha_yatri (     Booking_ID VARCHAR(20) PRIMARY KEY,     Date TIMESTAMP,     Time TIME,     Booking_Status VARCHAR(50),     Customer_ID VARCHAR(20),     Vehicle_Type VARCHAR(30),     Pickup_Location VARCHAR(100),     Drop_Location VARCHAR(100),     V_TAT INT,     C_TAT INT,     Canceled_Rides_by_Customer VARCHAR(255),     Canceled_Rides_by_Driver VARCHAR(255),     Incomplete_Rides VARCHAR(10),     Incomplete_Rides_Reason VARCHAR(255),     Booking_Value DECIMAL(10,2),     Payment_Method VARCHAR(30),     Ride_Distance INT,     Driver_Ratings DECIMAL(3,2),     Customer_Rating DECIMAL(3,2) ); ALTER TABLE ola RENAME TO odisha_yatri;  -- Q1: All successful bookings SELECT * FROM odisha_yatri WHERE booking_status='Success'; -- Q2: Avg ride distance per vehicle type SELECT Vehicle_Type, AVG(Ride_Distance) FROM odisha_yatri GROUP BY vehicle_type; -- Q3: Customer cancellation count SELECT COUNT(*) FROM odisha_yatri WHERE booking_status='Canceled by Customer'; -- Q4: Top 5 customers by bookings SELECT customer_id, COUNT(booking_id) AS total_ride FROM odisha_yatri GROUP BY customer_id ORDER BY total_ride DESC LIMIT 5; -- Q5: Driver cancellations - personal/car SELECT COUNT(*) FROM odisha_yatri WHERE canceled_rides_by_driver='Personal & Car related issue'; -- Q6: Prime Sedan rating range SELECT MAX(Driver_Ratings), MIN(Driver_Ratings) FROM odisha_yatri WHERE Vehicle_Type='Prime Sedan'; -- Q7: UPI payment rides SELECT * FROM odisha_yatri WHERE payment_method='UPI'; -- Q8: Avg customer rating per vehicle type SELECT vehicle_type, AVG(customer_rating) FROM odisha_yatri GROUP BY vehicle_type; -- Q9: Total revenue from successful rides SELECT SUM(booking_value) FROM odisha_yatri WHERE booking_status='Success'; -- Q10: Incomplete rides SELECT booking_id, incomplete_rides_reason FROM odisha_yatri WHERE incomplete_rides='Yes';
```

Appendix B: Dataset Summary Statistics

Statistic	Value
Total Raw Records	1,03,024
Total Unique Records (Post-Clean)	20,408
Duplicate Records Removed	82,616
Total Revenue (Success rides)	₹35,080,467
Success Rate	62.09%
Cancellation Rate	28.08%
Incomplete Rides	3,926
Vehicle Types	7
Payment Methods	4
Date Range	July 1–31, 2024

Statistic	Value
Max Driver Rating (Prime Sedan)	5.00
Min Driver Rating (Prime Sedan)	0.00
Top Customer (Revenue)	CID785112 — ₹8,025
UPI Transactions	25,881
Driver Cancellations (Personal/Car)	6,542

Appendix C: CTTC Course Details

Subject	Coverage
Introduction to Data Analytics and Python	Fundamentals of analytics, Python basics, libraries
Data Analysis using Python	Pandas, NumPy, Matplotlib, data wrangling
Machine Learning Models	Supervised & unsupervised learning, scikit-learn
SQL	DDL, DML, DQL, aggregate functions, joins
Advance Excel	PivotTables, Power Query, data cleaning, formulas
Tableau / Power BI	Dashboard design, calculated fields, story points

Appendix D: Glossary of Terms

Term	Definition
Booking_ID	Unique alphanumeric identifier assigned to each ride booking
Booking_Status	Final outcome of a booking: Success, Canceled by Customer, Canceled by Driver, or Driver Not Found
V_TAT	Vehicle Turn Around Time — time (in seconds) for the driver to reach the customer
C_TAT	Customer Turn Around Time — time (in seconds) for the customer to board after driver arrival

Term	Definition
DAX	Data Analysis Expressions — formula language used in Microsoft Power BI for calculations
CRISP-DM	Cross-Industry Standard Process for Data Mining — industry-standard analytics methodology
KPI	Key Performance Indicator — measurable value demonstrating effectiveness of objectives
EDA	Exploratory Data Analysis — statistical summarization to discover patterns in data
Power Query	Data connection and transformation component of Microsoft Power BI
UPI	Unified Payments Interface — India's real-time payment system
INR	Indian Rupee — the currency unit used for all revenue figures in this project
CTTC	Central Tool Room and Training Centre — the Government of India institution hosting this internship

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